2026 MCM Senior Group | Problem A

SafeInvest: Horizon-Aware Allocation with Cost-Gated Portfolio Updates

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Abstract

SafeInvest must choose suitable investments, avoid unnecessary trading, and recognize when automated advice should stop. We develop three horizon-specific portfolios using a convex expected-log-growth problem constrained by tail log drawdown. A separate execution rule moves partially toward the strategic allocation only when the estimated benefit exceeds transaction costs and a safety margin. Suitability is assessed at each customer's actual horizon using out-of-fold loss outcomes, conservative probability bounds, and a doubled-cost stress test. The selected, frozen Plan A assigns customers B and C to Conservative, E to Balanced, and A and D to human review.

All development uses 120 complete scenarios. A single frozen evaluation uses the remaining 60; its saved outputs provide the evidence reported here. At 30 years and a 0.25% transaction cost, Balanced A delivers a median real annual growth of 4.15%, compared with 3.38% for annual equal-weight rebalancing. Its 90th-percentile maximum drawdown is also higher: 48.02% versus 39.53%. Against annual rebalancing to the same strategic allocation, the execution rule reduces mean real fees by 26.83% and trades at 35.34% of subsequent annual reviews. A severe scenario nevertheless loses 65.19% from peak to trough. The evidence supports selective, explainable automation with explicit escalation; it does not establish a guarantee against loss or universal superiority over the baseline.

1 Decision Problem and Evidence

The advisor invests in equities, government bonds, and Treasury bills. Its responsibilities are linked: define Conservative, Balanced, and Growth products; match customers to a product or a human advisor; and choose among keeping holdings, rebalancing, changing the current target, and requesting human review. Optimizing returns alone does not address these service decisions.

The supplied data comprise 180 independent synthetic scenarios, each with 30 annual observations of asset returns and inflation. They are based on selected US series from the JST Macrohistory Database, Release 6, as described in the problem statement [1, 4]. Scenario numbers identify simulated paths, not historical episodes. We retain each complete path as one statistical unit and use the first H periods when evaluating a customer with horizon H.

Scenarios 1–120 form the development partition; scenarios 121–180 are reserved for the final evaluation. The latter was executed once on 17 October 2026 after model settings, customer assignments, and comparators were frozen. The final decision retains calibrated Plan A because subsequent development candidates did not satisfy the complete replacement criteria. Neural-network experiments are therefore outside the submitted strategy. Report preparation uses saved summaries and verification records only.

Our design separates three questions: what long-run allocation is appropriate, whether the customer can be served automatically, and whether an immediate trade is worth its cost. This makes the recommendation auditable and permits a justified refusal to automate.

2 Accounting, Assumptions, and Performance Measures

2.1 A causal, self-contained investment account

Initial nominal wealth is 100, held entirely in bills. Portfolios are long-only and fully invested: $w_i \geq 0$ and $\sum_i w_i = 1$. There are no contributions, withdrawals, leverage, taxes, or other fees. Annual decisions use only information available before that year's returns. Initial investment incurs a transaction fee; there is no assumed terminal liquidation. Bills provide liquidity but are not guaranteed to preserve purchasing power.

Let $h_{i,t}^-$ be pre-trade nominal holdings, $V_t^- = \sum_i h_{i,t}^-$, and $p_{i,t} = h_{i,t}^-/V_t^-$. A target w_t moves the following amount and incurs the following fee at rate c:

$$Q_t = \frac{V_t^-}{2}\|w_t - p_t\|_1, \qquad F_t = cQ_t, \qquad h_{i,t}^+ = (V_t^- - F_t)w_{i,t}. \tag{1}$$

The one-half factor counts the transferred money once, rather than charging separately for both sale and purchase. Turnover is measured against pre-fee wealth; the fee is deducted before returns, and the remaining wealth is allocated to the target. This states the accounting convention explicitly rather than assuming an implicit exact-dollar trade solver.

For nominal return $r_{i,t}$ and inflation π_t, define real return $\rho_{i,t}$, cumulative price index I_t, and year-end real wealth W_t by

$$\rho_{i,t} = \frac{1+r_{i,t}}{1+\pi_t} - 1, \qquad I_t = \prod_{k=1}^{t}(1+\pi_k), \qquad W_t = \frac{\sum_i h_{i,t}^+(1+r_{i,t})}{I_t}. \tag{2}$$

Pending human review means that existing assets remain invested and automatic trades stop. No unobserved human intervention, liquidation, or recovery is credited to the algorithm.

2.2 Growth, drawdown, and the cost of intervention

For a scenario with horizon H, real compound annual growth is

$$G = \left(\frac{W_H}{100}\right)^{1/H} - 1. \tag{3}$$

Maximum drawdown measures loss from a previous real-wealth peak. Let Z_j contain the initial wealth, each post-fee/pre-return wealth mark, and each year-end wealth mark. Then

$$D = \max_j \left(1 - \frac{Z_j}{\max_{k\leq j} Z_k}\right). \tag{4}$$

Including the post-fee marks ensures that trading losses are not hidden between annual observations. A larger D is worse. A stated customer tolerance L is evaluated through the event $D > L$; it is a preference threshold, not a guaranteed loss limit.

Across scenarios, we report median G, P10(G), median D, and P90(D). Annual volatility is the sample standard deviation of net real annual returns; its cross-scenario median summarizes typical variability. Terminal-loss probability measures $W_H < 100$. Drawdown CVaR at 90% averages the worst 10% of scenario drawdowns, with fractional boundary weights when needed; it is distinct from a 90th percentile.

The main trading frequency is the fraction of reviews in periods $2,\ldots,H$ that execute a trade. Initial allocation is excluded because it is compulsory; the alternative rate including it is reported when relevant. Mean real fees accumulate inflation-adjusted fees over the horizon, in units of initial wealth 100. Review incidence is the fraction of scenarios that ever request human review.

3 Strategic Portfolios and Customer Suitability

3.1 A convex allocation problem for each horizon

For N training paths and a fixed annually restored target w, define the largest cumulative log loss over any contiguous interval:

$$Y_s(w) = \max_{0 \le u \le v \le H} \left\{ - \sum_{t=u+1}^{v} \log(1 + \rho_{s,t}^{\mathsf{T}} w) \right\}. \tag{5}$$

The empty interval contributes zero. Under this cost-free, fixed-target model, ordinary maximum drawdown satisfies $D_s = 1 - \exp(-Y_s)$. Tail expectations must still be calculated on the intended variable: transforming $\mathrm{ES}(Y)$ does not generally give $\mathrm{ES}(D)$.

Using the variational representation of expected shortfall [2],

$$\mathrm{ES}_\alpha(Y) = \min_z \left[z + \frac{1}{N(1-\alpha)} \sum_{s=1}^{N} (Y_s - z)_+ \right], \quad \alpha = 0.90, \tag{6}$$

we choose the strategic anchor w^* by solving

$$\min_w \quad - \frac{1}{NH} \sum_{s=1}^{N} \sum_{t=1}^{H} \log(1 + \rho_{s,t}^{\mathsf{T}} w) \tag{7}$$

$$\text{subject to} \quad \mathrm{ES}_{0.90}(Y(w)) \le q R_{\mathrm{ref}}, \quad w \ge 0, \quad \mathbf{1}^{\mathsf{T}} w = 1. \tag{8}$$

Conservative additionally requires $w_{\mathrm{bill}} \ge 0.40$. For each horizon, R_{ref} is the training log-drawdown expected shortfall of a reference allocation. The equity/bond/bill references are $(.20, .30, .50)$ for Conservative, $(.50, .35, .15)$ for Balanced, and $(.75, .20, .05)$ for Growth. Thus products differ through their risk budgets as well as the Conservative liquidity floor.

Negative log growth and the maximum of interval log losses are convex on the valid-return domain. Equation (8) is therefore a convex strategic subproblem. A Kelley cutting-plane linear-programming method uses supporting planes, a 5% simplex grid for initialization, and lower/upper objective bounds to assess convergence. The configured optimality-gap tolerance is 10^{-6}, risk tolerance is 10^{-8}, and iteration cap is 300. Infeasibility is recorded rather than repaired by silently relaxing the risk budget. These certificates concern the strategic subproblem; they do not prove global optimality of the later trading and escalation system.

3.2 Development selection before freezing

Nested validation uses five outer scenario folds and three inner folds, preserving complete paths. Outer training/validation sizes are 96/24; inner training/validation sizes are 64/32. We compare $q \in \{.75, .85, .95\}$ and execution curvature $\gamma \in \{4, 12\}$, using held-out development outcomes rather than training fit alone. This separation addresses selection bias from repeated model comparisons [3].

Inner feasibility checks cover strategic feasibility, pooled log-drawdown risk, growth relative to same-anchor annual rebalancing, and review frequency. The validation risk budget is fixed independently of the candidate q; the risk slack is 0.005. Permitted median/P10 annual-growth shortfalls are 0.001/0.002, and review incidence is capped at 10%. Feasible candidates maximize mean $\log(1+G)$; a minimum-violation fallback is explicitly diagnostic. The selected full-development refit uses $q = 0.95$ and $\gamma = 4$ for all three products, then freezes the anchors and customer assignments before final evaluation.

3.3 Frozen allocations and an explicit refusal option

Table 1 shows that a product name does not imply one allocation for every customer. Shorter-horizon Balanced portfolios hold more bills, while the 30-year Growth solution reaches the boundary with zero bills. These are model outputs, not general investment prescriptions.

Table 1: Frozen strategic equity / bond / bill weights (%). Rounding can make totals differ slightly from 100.

H	Conservative	Balanced	Growth
5	24.48 / 4.46 / 71.05	54.26 / 0.00 / 45.74	75.21 / 0.00 / 24.79
6	24.22 / 1.59 / 74.20	53.25 / 4.46 / 42.29	74.61 / 0.00 / 25.39
15	22.67 / 7.23 / 70.10	48.65 / 21.78 / 29.58	71.38 / 23.59 / 5.03
25	25.98 / 2.84 / 71.18	49.78 / 18.43 / 31.79	71.24 / 24.35 / 4.41
30	26.48 / 2.85 / 70.67	50.81 / 14.44 / 34.75	70.91 / 29.09 / 0.00

For customer horizon H and tolerance L, we count out-of-fold development paths with $D > L$, using the complete execution policy. If k of $n = 120$ paths breach, the one-sided Clopper–Pearson upper bound is

$$U = \text{Beta}^{-1}(1-\delta; k+1, n-k), \qquad \delta = 0.05/440, \tag{9}$$

with $U = 1$ when $k = n$. The 440-comparison family is the prespecified service-screening budget. Automatic service requires $U \leq 0.20$ under both the base cost and doubled cost. Because fitted out-of-fold policies share training observations, this is a conservative screening diagnostic; it is not a strict coverage guarantee for the final refitted policy.

High-liquidity customers require a strategic bill weight of at least 40% in every relevant development fold. This condition is checked for any eligible product, rather than restricting high-liquidity customers to the product named Conservative. It is a target requirement, not a continuously maintained cash floor after market drift. Growth is available only for horizons of at least 15 years. We also check that P90 drawdown is ordered Conservative $\leq$ Balanced $\leq$ Growth at the relevant horizon and costs. Among eligible products, the highest out-of-fold median growth is selected; if none qualify, the customer goes to human review.

Table 2: Customer requirements and assignments frozen before final evaluation.

Customer	H	Tolerance (%)	Liquidity	Assignment
A	5	10.00	High	Human review
B	15	25.00	Medium	Conservative
C	30	40.00	Low	Conservative
D	25	15.00	Medium	Human review
E	6	35.00	High	Balanced

B and C qualify for Conservative: their base-cost breach counts are 3/120 and 0/120, with adjusted upper bounds 12.45% and 7.29%. C's Balanced alternative has an upper bound of 28.93%, exceeding the service ceiling despite a raw breach rate of 14.17%. E qualifies for Balanced at six years: its bound is 13.87% at both costs, and its minimum fold-level bill target is 40.03%. A and D fail even the Conservative risk screen, with base upper bounds of 39.70% and 65.17%. Human review is therefore a substantive outcome of the model.

4 Annual Execution and Human Escalation

4.1 A small, explainable set of candidate trades

At an annual review, let p be the current portfolio and w^* its frozen horizon-specific strategic anchor. The advisor considers

$$w_\lambda = p + \lambda(w^* - p), \qquad \lambda \in \{0, .25, .50, .75, 1\}. \tag{10}$$

Training data supply the real-return mean μ and a regularized covariance matrix $\Sigma = 0.75S + 0.25\,\mathrm{diag}(S) + 10^{-10}I$. Define a local tracking score

$$A(w) = 1 + \mu^\top w^* - \frac{\gamma}{2}(w - w^*)^\top \Sigma (w - w^*), \qquad f_\lambda = \frac{c}{2}\|w_\lambda - p\|_1. \tag{11}$$

The expected-return term uses the strategic anchor; this penalizes departure from a chosen allocation, rather than predicting each candidate's return. If any candidate score is nonpositive or nonfinite, the advisor requests human review. Otherwise, select j maximizing $(1 - f_\lambda)A(w_\lambda)$. A positive-size trade is accepted only if

$$\frac{A(w_j) - A(p)}{A(w_j)} > f_j + 0.0001. \tag{12}$$

The additional 0.0001 is a one-basis-point safety margin. Candidate ranking precedes this strict gate; a rejected candidate leads to keeping holdings. The heuristic controls trading frequency but does not certify an improvement in realized wealth on every future path.

4.2 Four observable actions

The decision process first checks the observed drawdown and pending-review status, then evaluates the trade candidates:

1. **Keep holdings:** no candidate passes the cost gate, or the chosen step is zero. No fee is charged.
2. **Rebalance:** an accepted trade restores the previously recorded target.
3. **Change the current target:** an accepted partial adjustment produces a different execution target. The strategic anchor remains fixed.
4. **Request human review:** the loss trigger or an invalid decision score makes continued automatic trading inappropriate. The engine keeps existing assets invested while review is pending.

Before acting, the advisor requests review if observed real drawdown is at least

$$\tau = \mathrm{clip}(1.5\ \mathrm{P90}(D_{\text{train,annual}}),\ 0.10,\ 0.75), \tag{13}$$

where the training drawdowns come from fee-paying annual rebalancing to the same anchor. The 30-year Balanced threshold is 63.85%. A trigger based on an annual observation cannot prevent a large loss during the following year, and a request for review does not itself move wealth to safety.

4.3 Customer-facing explanation

Each accepted adjustment can be explained in the following 20 words, within the required 50-word limit:

> "This adjustment moves your investments toward the strategic allocation. Its estimated risk-adjusted benefit exceeds trading costs. Investment losses remain possible."

The explanation states the decision criterion and remaining uncertainty. It avoids promising an improved realized return or recovery of previous losses.

5 Results from the Single Frozen Final Evaluation

5.1 Thirty-year performance and fair comparisons

Table 3 uses the same 60 final scenarios and a fee of 0.25% of money moved. Equal-weight annual rebalancing is the required baseline. Same-anchor annual and hold controls isolate execution from allocation; both use the frozen Balanced anchor. Median annual volatility is 10.76% for Balanced A and 8.24% for equal weight.

Table 3: Thirty-year final outcomes (%). *G*: real annual growth; *D*: maximum real drawdown. Trade rate excludes initial allocation.

Policy	Med. *G*	P10 *G*	Med. *D*	P90 *D*	Trades
Equal-weight annual	3.38	1.50	22.30	39.53	100.00
Balanced A	4.15	1.64	30.58	48.02	35.34
Same-anchor annual	4.02	1.67	30.30	48.12	100.00
Same-anchor hold	4.27	1.42	36.05	59.15	0.00
Conservative A	2.76	1.35	19.19	31.91	28.45
Growth A	5.14	2.07	38.23	62.15	28.56

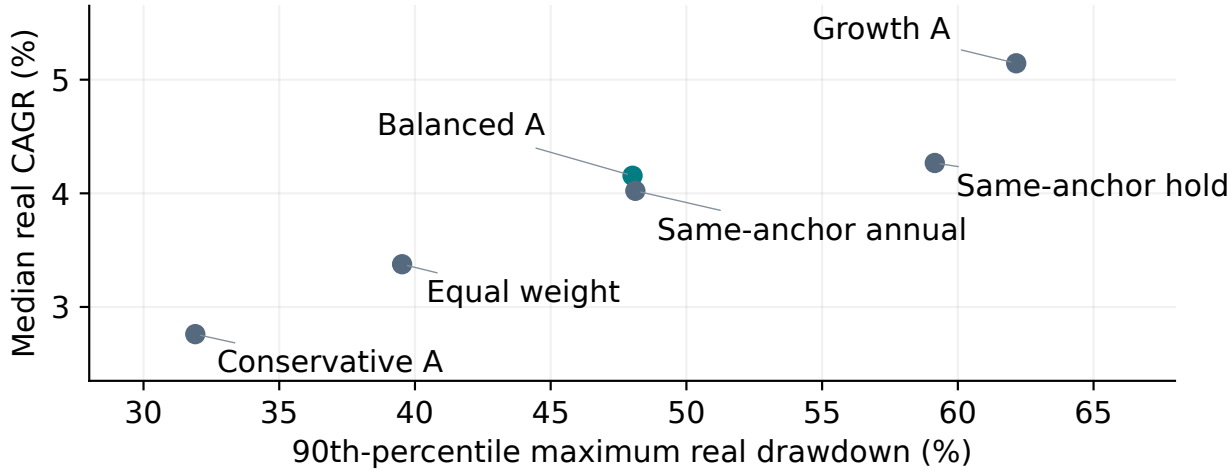


Figure 1: The selected policy improves median growth relative to equal weight while accepting greater tail drawdown. Neither axis alone establishes suitability.

Balanced A gains 0.78 percentage points (pp) of median annual growth over equal weight, but its P90 drawdown increases by 8.49 pp. Saved paired bootstrap intervals, using 4,000 resamples of complete scenarios, give a 95% interval of [0.325, 1.184] pp for the median-growth difference and [5.203, 13.365] pp for the P90-drawdown difference. Family-adjusted intervals are [0.098, 1.372] and [3.399, 15.735] pp, respectively. The adjustment covers 20 nonbaseline policies separately for each endpoint, horizon, and cost; it does not cover all strata or all adaptive research.

Against same-anchor annual rebalancing, Balanced A has 0.13 pp higher median growth and nearly unchanged P90 loss. Mean real fees fall from 0.7973 to 0.5834 (a 26.83% reduction). The trading rate is 35.34%, or 37.50% when initial allocation is included. Relative to holding the anchor without rebalancing, it reduces P90 loss by 11.13 pp, at a median-growth cost of 0.11 pp. These fixed-comparator results support the cost gate without claiming universal dominance.

5.2 Customer outcomes at their actual horizons

The 30-year product comparison must not be substituted for individual customer evidence. Table 4 evaluates the already frozen assignments at each actual horizon. A and D have no automated investment path; missing outcomes are not zero losses. Breaches are counted against each customer's stated tolerance.

Table 4: Final customer outcomes (growth and drawdown in %; 60 scenarios).

Customer	H	Med. G	P10 G	Med. D	P90 D	Breaches
A	5	–	–	–	–	–
B	15	2.94	0.48	12.57	25.38	7/60
C	30	2.76	1.35	19.19	31.91	3/60
D	25	–	–	–	–	–
E	6	4.11	-0.35	11.95	26.89	3/60

Table 5: Additional customer risk and intervention measures. Terminal loss means final real wealth below 100.

Customer	Median volatility (%)	Terminal losses	Trade rate (%)
B	6.26	3/60	22.86
C	6.75	3/60	28.45
E	10.08	9/60	30.67

For B, P90 drawdown is slightly above the 25% preference threshold and 7 of 60 scenarios breach it. This is compatible with a service rule based on breach probability rather than a deterministic cap. For E, positive median growth coexists with a negative P10 growth and 9 terminal-loss scenarios. B, C, and E request human review in 2, 3, and 0 final scenarios, respectively. These observations are disclosed outcomes, not grounds for changing the assignments after seeing the holdout.

5.3 A severe path exposes the limits of escalation

Scenario 155 is the saved severe example for 30-year Balanced A. Table 6 records its observed milestones. These values document the sequence of peak wealth, escalation, and terminal loss without interpolating an annual wealth path.

Table 6: Severe scenario 155: saved milestones, initial real wealth 100.

Measure	Observed value
Real-wealth peak	162.33 at the end of year 6
Human-review request	Before the year-23 return
Trough and terminal real wealth	56.50 at the end of year 30
Maximum drawdown	65.19%
Real annual growth	-1.88%
Subsequent reviews with trades	6

The peak-to-trough loss exceeds the 63.85% escalation threshold because review is neither continuous nor an executed hedge. Holdings remain exposed while human review is pending. Across all 60 Balanced paths, drawdown CVaR90 is 59.31%, compared with P90 drawdown of 48.02%. This tail evidence argues for candid risk disclosure and for distinguishing an escalation mechanism from capital protection.

6 Sensitivity and Reliability

6.1 Transaction-cost stress with the policy frozen

We vary fees across five rates on the same 60 final scenarios, retaining the frozen strategic parameters and customer assignments. The cost gate responds to each rate (Table 7, Figure 2).

Table 7: Thirty-year Balanced A, 60 final scenarios. Growth and drawdown are in %; mean fees are cumulative real wealth units (initial wealth 100). Trade rates exclude initial allocation.

Cost (%)	Med. G	P10 G	P90 D	Mean fees	Trades (%)
0.000	4.08	1.68	47.96	0.0000	46.61
0.125	4.19	1.70	47.56	0.3199	39.60
0.250	4.15	1.64	48.02	0.5834	35.34
0.500	4.13	1.61	47.35	1.0668	28.68
1.000	4.13	1.61	47.72	1.8687	19.83

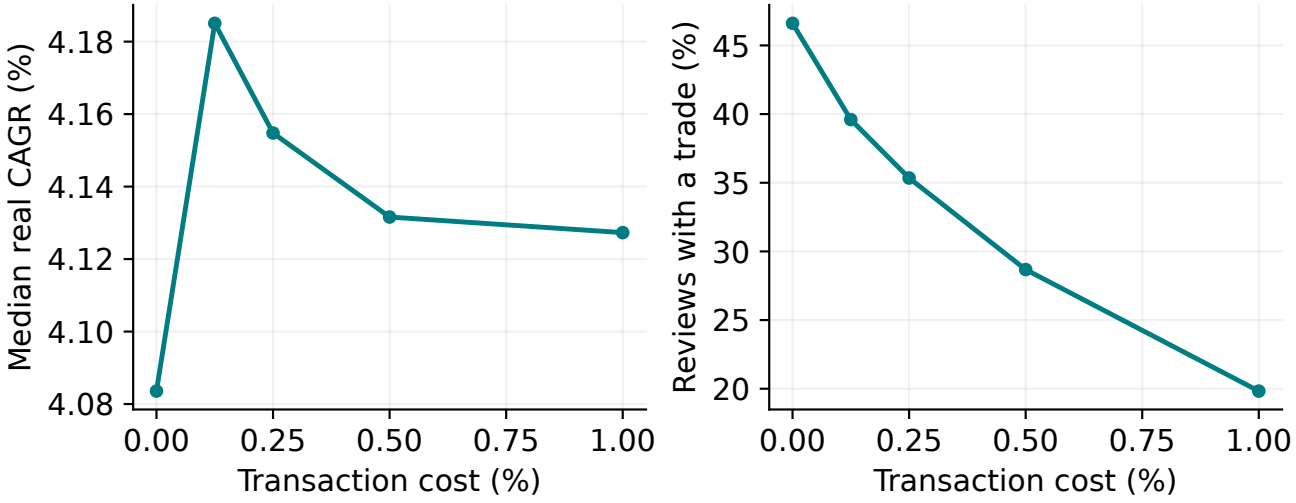


Figure 2: Trading decreases across the tested fee rates. Growth is nonmonotonic because changing fees also changes the trades.

Doubling the fee to 0.50% reduces trading from 35.34% to 28.68%; median growth changes from 4.15% to 4.13%, and mean real fees rise from 0.5834 to 1.0668. Nonmonotonic growth reflects changed exposures, not a benefit from fees.

For eligible customers, selective Balanced updates remain our recommendation: at every tested rate, median growth and P90 drawdown exceed equal weight. Customer eligibility was screened only at base and doubled costs during development.

6.2 What is verified, and what remains uncertain

Saved checks covered 286,920 transitions, 780 equal-weight scalar-reference cases, and five customer assignments. Maximum absolute reconstructed year-end holding error was 3.64×10^{-12} nominal wealth units. These checks support accounting consistency, not predictive reliability.

Synthetic data and only 60 independent final paths limit generalization and tail-risk precision; testing five fees does not add independent paths. Annual reviews miss within-year drawdowns, and the cost-free strategic risk proxy differs from realized execution risk. Bill weights can drift below targets. Withdrawals, taxes, additional frictions, and effective human intervention are unmodeled.

7 Recommendation and Conclusions

We recommend the frozen calibrated Plan A. Conservative uses the lowest reference risk budget and a strategic bill floor of 40%; Balanced uses an intermediate budget; Growth uses the highest budget and is eligible only for horizons of at least 15 years. Their horizon-specific allocations are given in Table 1. Applying the suitability screen at each customer's horizon assigns B and C to Conservative, E to Balanced, and A and D to human review.

Balanced should use selective automated updates when estimated benefit exceeds transaction costs and the safety margin. At 30 years and the base fee, this cuts mean real fees by 26.83% relative to annual rebalancing to the same anchor, with similar P90 drawdown. Against equal weight, median growth is higher but tail drawdown is worse. This supports cost-aware execution for suitable customers, without establishing universal risk or return superiority.

The main safeguard combines refusal at onboarding with suspension of automatic trades when observed real drawdown reaches the training-derived review threshold or model scores are invalid. Existing assets remain invested pending human review. For 30-year Balanced, the threshold is 63.85%, and the severe path still loses 65.19% from peak to trough. Escalation therefore cannot be presented as capital protection.

Reproducibility and attribution

The frozen model, development selection record, one-time final-evaluation lock, and comparator outputs preserve the evaluation trail. Report tables and figures use these saved records. Supplementary source code is submitted separately.

The problem and supplied synthetic data are credited to the 2026 MCM Senior Group organizers and the stated underlying JST source [1, 4]. We acknowledge their CC BY-NC-SA 4.0 license; derived data displays retain this attribution and the same terms. Reported returns describe the synthetic experiment and are not forecasts.

References

[1] 2026 MCM Senior Group Organizers. *Problem A: SafeInvest: Building a Trustworthy Robo-Advisor*. Competition problem statement and supplied scenario data, 2026.

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Use of AI Tools

OpenAI ChatGPT/Codex assisted with code development, analysis, drafting, and document checks. Numerical claims in this report are drawn from saved experiment outputs and frozen model settings; the report-building step did not refit a model or rerun the final evaluation. AI-generated explanations and prose require review by the named team members before submission. The authors remain responsible for the correctness, originality, attribution, and final submitted content.